

On Predicting and Analyzing Breast Cancer using Data Mining Approach

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6/8

Abstract—The highest invading cancer among the women is breast cancer. Early detection of breast cancer is the higher chance of the patient being treated. In this study, we have proposed an ensemble method named stacking classifier which combines multiple classification techniques and efficaciously classifies the benign and malignant tumor. “Wisconsin Diagnosis Breast Cancer” dataset culled from the UC Irvine Machine Learning Repository has been used for our experiment. We applied different classification techniques over the dataset and tuned their parameters to improve accuracy. We chose the three best classifiers for our proposed method. Generally, our proposed Stacking classifier combined the results of those best classifiers using meta classifier and provided 97.20% accuracy for breast cancer prediction. Performance of different data mining approaches have been evaluated rigorously through different evaluation metrics.

Keywords—This breast cancer prediction, ensemble method, stacking classifier.

I. INTRODUCTION

The cancerous agent which affects the tissue of the breast is identified as breast cancer. Now it is one of the crucial and burning disease that has invaded women. Changing of breast’s skin, dumpling skin, breast pain, swelling underarm, changing of breast shape, and inverted nipple are the signs of breast cancer [1]. Approximately 5-10% of this disease occurrences happen the reason for ancestral, and attack among the ladies by changing BRCA1 & 2 [2]. Data mining is the process of running powerful classification techniques that extract useful information from data. The uses and potentials of these techniques have found its scope in medical data. Classification techniques tend to simplify the prediction segment. Generally, it is very hard and time-consuming to classify breast cancer for the doctor. So, predicting cancer through various automatic diagnostic techniques

is very necessary. This study proposes an ensemble method, namely staking classifier for predicting breast cancer. Stacking Classifier is used to merge different classification techniques with the help of meta-classifier. Every technique is trained on a dataset and provides different output. These outputs are provided as input into a meta-classifier. The meta-classifier provides final output by combining specific classification technique’s output. In this paper, “Wisconsin Diagnostic Breast Cancer” dataset is used where the dataset’s target class label has two categories of tumors as Benign and Malignant and Malignant tumor contains the cancerous agent. The contribution of this paper is to show that the ensemble method, namely stacking classifier can correctly classify between malignant and benign tumors. The objectives of this paper are as follows:

- To predict and analyze the Breast Cancer using different classification techniques.
- To preprocess the dataset and select top suitable features from the dataset which are enough to classify the Breast cancer.
- To train different classifier techniques on the processed dataset.
- To analyze performance of different classification techniques and build the best model based on their performance.

In this paper, Section 2 narrates a list of previous research works. Section 3 shows the workflow of study and section 4 analyses the proposed outline. Section 5 displays the experimental outcome. At last, the paper has been concluded in section 6.

II. LITERATURE REVIEW

In this section, we highlighted some of the research works previously done over Breast Cancer prediction. Generally, most of the researchers have worked in breast

cancer using the Wisconsin dataset, SEER dataset, and also dataset from different hospitals. R. Nithya et al. investigated and examined the classification technique multiboostSMO for classifying breast cancer [3]. Shubham Sharma et al. applied Random Forest, KNN, and Naïve Bayes for breast cancer prediction [4]. Shang Gao et al. used the SPSS Clementine data mining tool and analyzed with various kernel functions and parameters of the SVM [5]. Naresh Khuriwal et al. explained the performance of ANN approach and logistic algorithm with an ensemble machine learning algorithm [6]. Mohamed NEMISSI et al. introduced a neural classification system with a single hidden neural network and compared to the conventional extreme learning network [7]. Mümine KAYA KELEŞ applied Bagging, IBk, Random Forest, SimpleCART, and Random Committee classification techniques for breast cancer prediction and also applied UWB radar-based systems [8]. Zarei.S et al. analyzed KNN, logistic regression, and multivariate linear regression and classified tumor type on Wisconsin dataset [9]. Vikas Chaurasia et al. applied the best tree, IBK and SMO to increase classification accuracy [10].

Our paper is different from the previous research papers in the sense that we have combined multiple classifier techniques using a meta classifier with the help of Stacking classifier. Generally, the Stacking classifier is an ensemble method that provides high performance.

III. PROPOSED DIAGRAM

In this section, we have been shown our workflow to predict breast cancer.

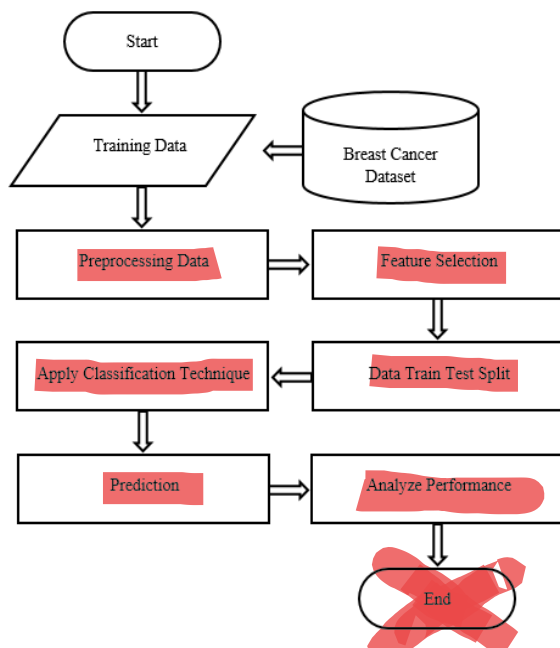


Fig. 1. Flow Diagram for Breast Cancer Prediction.

In Fig. 1, firstly culled the Wisconsin breast cancer dataset from the UC Irvine Machine Learning

Repository. After processing the dataset and choosing 20 top features using the Univariate Feature Selection method. Then shear the dataset in order to train and test. After that applied different classification techniques and predicted the result. Finally analyzed performance by different evaluation metrics.

IV. PROPOSED METHODOLOGY

This section describes our proposed methodology to predict the Breast Cancer.

A. Dataset Information

“Wisconsin Diagnostic Breast Cancer (WDBC)” dataset has 569 observations with 32 attributes. Features are extracted from a digitized image of fine needle aspirate (FNA) from a breast mass and that indicates the properties of cell nuclei. The class label contains 2 categories as Malignant and Benign. In between the class, 212 observations are malignant and 357 observations are benign.

B. Dataset Preprocessing

In this study, Wisconsin dataset has contained classes as M and B depicting malignant and benign respectively. Categorical values are converted into numeric values using label encoding. In this paper, we are removed outliers using interquartile range (IQR) method and also applied to a standard scaling method to normalize the data.

C. Feature Selection

Using the Univariate Feature Selection method, we have selected 20 top features for breast cancer prediction. The method calculated the ch_2 score for each feature using the ch_2 formula. The feature which contains a high score is connoted for the target class.

D. Proposed Classification Technique

The classification techniques applied in this work are CART, Random Forest, Logistic Regression, K-Nearest Neighbors, Support Vector Machine. First of all, we have applied these techniques to the processed dataset for training and compute the accuracy. Then we are tuned to these techniques’ parameters using grid search cross-validation and generate the best parameter for these techniques. Again, we have applied these techniques with their best parameter on the dataset and compute the accuracy. After we compared their accuracy and selected the best four techniques for our proposed technique. In this paper, we have proposed a classification technique named Stacking Classifier that is an ensemble method. Stacking Classifier has two levels. In level 0, applying multiple classification techniques on the dataset with 10-fold cross-validation and compute each technique’s individual output. These results are provided as input in level 1. In level 1, applying a meta classifier that combines the previous output and provides a final output. Algorithm 1 shows the steps of the proposed classification technique.

Algorithm 1. Stacking Classifier

Input: Dataset, $D = \{X_i, y_i\}_{i=1}^m$; X = feature attribute, y = target class.
Output: H ; ensemble classifier

1. Partitioned dataset randomly, $D = \{D_1, D_2, \dots, D_k\}$ // k indicates number of folds.
2. **for** $k \leftarrow 1$ to k **do** // Level 0 classifiers
3. **for** $t \leftarrow 1$ to c **do** // c indicates number of classifiers
4. Train level 0 classifier h_{kc}
5. **end for**
6. **for** $X_i \in D_k$ **do** // New training set for level 1
7. Create new dataset $\{X_i^{new}, y_i\}$,
 where $X_i^{new} = \{h_{k1}(X_i), h_{k1}(X_i), \dots, h_{kc}(X_i)\}$
8. **end for**
9. **end for**
10. **for** $t \leftarrow 1$ to c **do** // Train a new classifier h^{new}
11. Train level 1 classifier h_c based on the new dataset
12. **end for**
13. **return** $H(X) = h^{new}(h_1(X), h^{new}(h_2(X), \dots, (h_c(X)))$

In Algorithm 1, firstly partitioned the dataset with k -fold cross-validation. In level 0, multiple classifiers are trained until k -times. The predicted output for different classifiers of level 0 is used as input in level 1. In level 1, a meta classifier is retrained as level 0 where the meta classifier combines previous output and provides the final result.

E. Performance Model Evaluation

The evaluation of data mining classification technique's performance involves testing the proposed model. **Confusion matrix** is used for evaluating the performance. Also, different performance evaluation parameters, namely **accuracy**, **ROC area**, **precision**, **recall**, **F1 score** which are applied to evaluate the performance of the classification technique.

V. EXPERIMENTAL RESULT ANALYSIS

To implement different classification techniques such as CART, Logistic Regression, KNN, SVM, Random Forest, and Stacking Classifier techniques, **Jupyter Notebook** is used with some **python** open source **library**.

We have applied k -fold cross-validation method for training and testing classification techniques. CART, Logistic Regression, KNN, SVM, Random Forest classification techniques are implemented on our processed dataset for training. Then we are tuned to these technique's parameters using **grid search cross-validation** with **10-fold** cross-validation and generate the best parameter for these techniques. Again, we have applied these techniques with their best parameter on our dataset and compute the accuracy. After we are compared their accuracy and we have got over 95% accuracy for Logistic Regression, K-Nearest Neighbor, Support Vector Machine, and Random Forest. Then we have chosen **K-Nearest Neighbor**, **Support Vector Machine**, and **Random Forest** as a base classifier and **Logistic Regression** as **meta classifier** for **proposed method Stacking Classifier**. Applying Stacking Classifier, we are combined with the base classifier's prediction by meta classifier Logistic Regression.

Fig. 2 shows the blending of decision boundaries achieved by the stacking classifier where 0 represents benign and 1 represents malignant tumor and we can see Stacking Classifier perfectly classifies benign and malignant tumor.

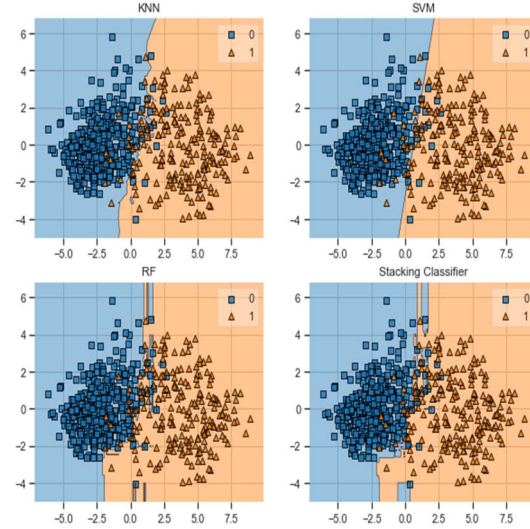


Fig. 2. Decision Boundary for Classification Technique.

Fig. 3 shows that the Stacking classifier provides the best accuracy of 97.20% after combining KNN, SVM, and Random Forest classification techniques.

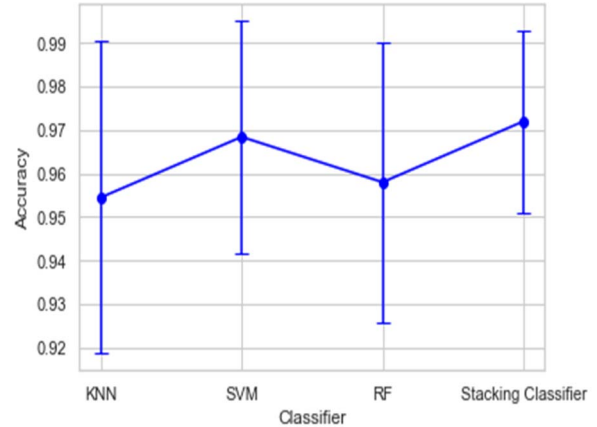


Fig. 3. Comparison of Accuracy.

The overall outcomes displayed in Table I which indicate Stacking Classifier has the highest accuracy 97.20%, precision 97.24%, recall 95.30% and F1 score 96.16% where CART has the lowest accuracy 94.74%, precision 92.12%, recall 93.65 and F1 score 92.91%. Also, we can see that **Stacking Classifier has the best Matthews Correlation Coefficient (MCC)** value 0.938 and ROC area 0.993 and lowest mean absolute error 0.028 and R2 score 0.880. So, it represents that Stacking Classifier provides much better performance from other classifiers. In Fig. 4, the result of different performance

evaluation parameters for all classification techniques is displayed.

VI. CONCLUSION AND FUTURE WORK

The automatic prediction of breast cancer is significant to abate the propensity against enlarging this disease. Early detection of breast cancer is the key to treatment. This study proposed a classification technique,

namely stacking classifier for breast cancer prediction and that achieved 97.20% accuracy. The percentage of the accuracy indicates that cancerous tumor can be detected ideally. We have selected 20 top features for the prediction of breast cancer. Furthermore, it is planned to collect data from various regions across the world and create a more accurate and stronger predictive model for the prophecy of breast cancer.

TABLE I. PERFORMANCE MEASURE INDICES

Model Performance									
Classifier	Accuracy (%)	Classification Error (%)	Precision (%)	Recall (%)	F1 Score (%)	MCC	ROC area	Mean absolute error	R2 score
CART	94.74	5.26	92.12	93.65	92.91	0.887	0.977	0.053	0.774
LR	97.08	2.92	93.93	98.41	96.12	0.938	0.998	0.292	0.874
KNN	95.91	4.09	95.16	93.65	94.40	0.912	0.992	0.041	0.824
SVM	95.91	4.09	95.16	93.65	94.40	0.912	0.997	0.041	0.824
RF	97.08	2.92	93.94	98.13	96.12	0.938	0.997	0.029	0.874
Stacking Classifier	97.20	2.80	97.24	95.30	96.16	0.938	0.993	0.028	0.880

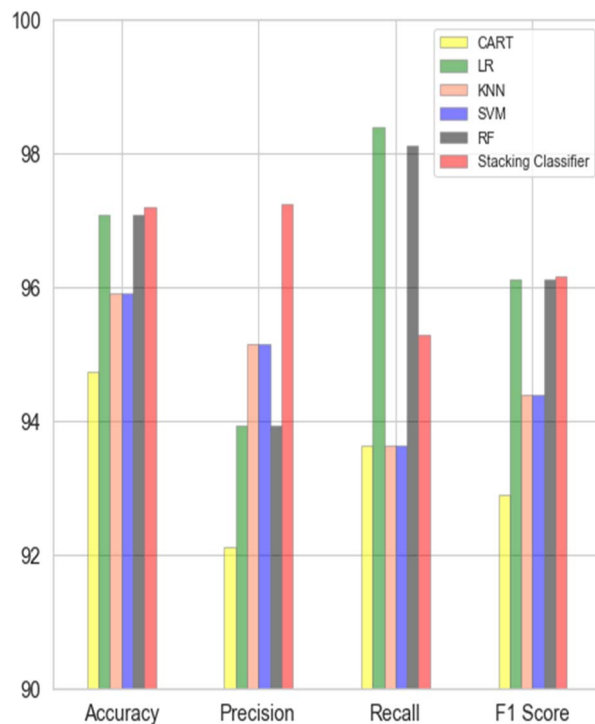


Fig. 4. Graphical Representation of Classifier Performance Measure.

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